RAGHAVAN PIRAPANCHAN

AI Engineer

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ABOUT ME

I am an AI undergraduate passionate about creating innovative solutions and improving intelligent systems. I excel at solving problems by breaking down complex issues and finding effective solutions. I enjoy working in teams, using my strong communication skills and flexible nature to help achieve project success.

EDUCATION

- **University of Moratuwa** (May 2022 – 2026)
  BSc(Hons) in Artificial Intelligence

- **Jaffna Hindu College** (2010 – 2018)
  G.C.E. Advanced Level Examination (2020)
  Combined Maths - A , Physics - C , Chemistry - C (Z-Score - 1.2001)

EXPERIENCE

- **Techorin** (March 2025 – September 2025)
  AI Engineer Intern

  Worked on designing and developing AI driven solutions focused on computer vision based detection systems. Collaborated with cross functional teams to implement vision models for real world applications and gained hands on experience in deploying and optimizing AI solutions.

TECHNICAL SKILLS

- Programming Languages **- Python, Java, C**
- Libraries **- NumPy, Pandas, TensorFlow, PyTorch, Scikit-Learn, Matplotlib**
- Web Development **- React Native, JavaScript, HTML, CSS**
- Backend **- FastAPI, NodeJS**
- Database **- MySQL, MongoDB**
- Mobile App Development **- NativeBase**
- Version Controlling **- Git**

PROJECTS

- **An Explainable Fuzzy Vision Transformer for Temporal Ergonomic Risk Prediction** (2026 - Ongoing)
  Final Year Research Project

  Conducting research on an AI based ergonomic risk assessment framework for workplace safety using computer vision, Temporal Vision Transformers (Temporal-ViT), and Explainable AI techniques. Developed an end to end pipeline for pose extraction, ergonomic feature computation, REBA based risk labelling, and temporal sequence modelling from workplace video data. Currently implementing a fuzzy inference reasoning layer to generate human interpretable ergonomic risk explanations aligned with industrial safety standards.

  Technologies – Python, TensorFlow, MediaPipe, OpenCV, NumPy, Pandas, Scikit-Learn, Scikit-Fuzzy, Streamlit

- **Financial Forecasting Tool Development** (2023-2024)
  2nd year Software project (Mentored by - Zone24x7)

  Developed a financial management tool to calculate and manage user balances, allowing budget planning based on cash and bank transactions. The system tracks daily expenditures and income, leveraging time series models for future financial predictions

  Technologies – React Native, NativeBase, MongoDB, FastAPI, Timeseries Model

- **ChatMate – AI-Powered Educational Chatbot** (2025 - Ongoing)
  Individual Project

  Developed a full stack AI chatbot application designed to enhance the education experience by providing intelligent query resolution and academic support. Built the frontend and backend using the MERN stack, implementing authentication and interactive chatbot pages. Currently integrating a LangChain based Retrieval Augmented Generation (RAG) model to improve chatbot responses with contextual and accurate information.

  Technologies – React, Node.js, MongoDB, Express.js, LangChain, RAG, OpenAI

- **Computer Vision-Based Ergonomic Posture Scoring** (2025)
  Individual Project

  Ergonomic assessment tool designed to analyze weightlifters posture while lifting. Evaluate joint strain and classify posture on a scale of 1-9 to determine whether movement is safe or risky. Using computer vision and deep learning, it processes video frames, detects human joints, and applies ergonomic guidelines to provide actionable feedback.

  Technologies – OpenCV, MediaPipe Pose, YOLOv5, PyTorch, NumPy

- **Shakespeare Text Generation Using LSTM Neural Networks**(Ongoing) (2024)
  Individual Project

  Developed an LSTM based text generation model capable of producing text in the style of Shakespeare. The project involved training the model to predict the next character in a sequence, enabling it to emulate the poetic structure and language of Shakespeare's works.

  Technologies – Python, TensorFlow, LSTM Neural Networks.

- **Carbon Footprint Reduction Expert System** (2024)
  Individual Project

  Developed a web-based expert system using Streamlit and Experta to assess carbon footprints in Energy Consumption, Transportation, and Waste Management, with a step by step questionnaire and rule based recommendations for reducing emissions.

  Technologies – Python, Streamlit, Experta

LANGUAGES

- English
- Tamil

OTHER SKILLS

- Teamwork
- Communication skills
- Creativity
- Problem solving

REFEREES

- **Dr. (Ms.) I. Thilini S. Piyatilake**
  Senior Lecturer, QA Coordinator
  Department of Computational Mathematics
  Faculty of Information Technology
  University of Moratuwa
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